

- 4 Alleles are alternative forms of a gene. For example, there may be a dominant allele, **T**, and a recessive allele, **t**, at the same gene locus.

The relative frequency of each allele of a gene in a population can change over time due to factors such as selection, genetic drift and the bottleneck effect.

- (a) Fig. 4.1 shows the relative frequency of the **T** allele in a population of 50 individuals over 20 generations.

At generation 0, the number of **T** alleles and the number of **t** alleles in the population was equal, so the relative frequency of each allele was 0.5. The relative frequencies of the two alleles of the gene add up to 1.

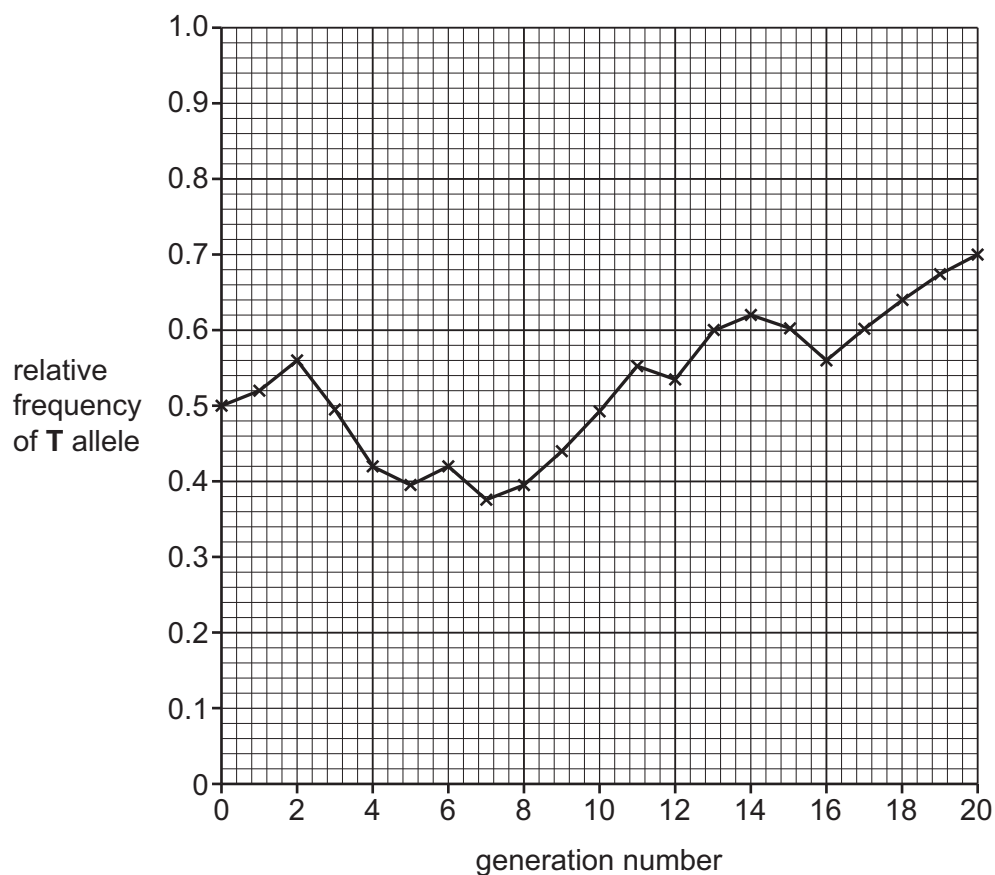
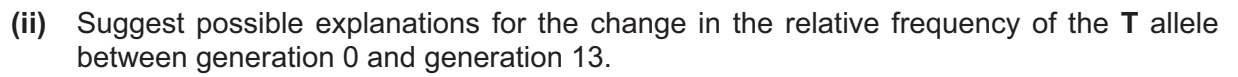


Fig. 4.1

- (i) With reference to Fig. 4.1, state the relative frequency of the **t** allele after 20 generations.

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[4]

(b) The relative frequency of the **T** allele in a population of only 10 individuals was determined over 20 generations. The environmental conditions remained the same throughout the experiment.

A line graph showing the relative frequency of the T allele over 20 generations. The y-axis is labeled 'relative frequency of T allele' and ranges from 0 to 1.0 in increments of 0.1. The x-axis is labeled 'generation number' and ranges from 0 to 20 in increments of 2. The data points are marked with asterisks and connected by a solid line. The frequency starts at 0.5 in generation 0, decreases to 0.4 in generation 2, then to 0.3 in generation 3. It rises to 0.45 in generation 4 and remains at 0.45 in generation 5. It then decreases to 0.4 in generation 6, 0.3 in generation 7, and finally drops to 0 in generation 8, where it remains for the rest of the 20 generations.

generation number	relative frequency of T allele
0	0.50
1	0.45
2	0.40
3	0.30
4	0.45
5	0.45
6	0.40
7	0.30
8	0.00
9	0.00
10	0.00
11	0.00
12	0.00
13	0.00
14	0.00
15	0.00
16	0.00
17	0.00
18	0.00
19	0.00
20	0.00

Explain why Fig. 4.2 shows a different result from Fig. 4.1.

[3]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32

- (c) State the name of the principle that can be used to calculate relative frequencies of two alleles by counting the numbers of organisms in the population showing dominant and recessive phenotypes.

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[Total: 9]